सेकी
SECI
सूर्य सदैव SUN FOR EVER
Solar Energy Corporation of India Limited (SECI)

POWERING A GREENER FUTURE

Solar Energy Corporation of India has ushered in a paradigm shift in India's energy sector, catalysing the transformation of renewable energy into a commercially viable and mainstream source of power

As India takes strides towards creation of 500 GW of installed electricity capacity from non-fossil sources by 2030 and net zero emissions by 2070, the Solar Energy Corporation of India Limited (SECI) emerges as a key protagonist in the unfolding narrative. Established in 2011 under the National Solar Mission (NSM), the organisation is dedicated to pioneering new models and solutions that meet India's evolving energy needs while driving economic growth and employment generation.

SECI serves as implementing agency of the Ministry of New and Renewable Energy (MNRE), tasked with the development of renewable energy (RE) projects on a pan-India scale. Backed by standard guidelines issued by the Union government, the establishment facilitates the procurement of power from developers, forging 25-year Power Purchase Agreements (PPAs) that ensure an assured off-take of clean energy.

But SECI's role extends far beyond mere facilitation. With a commitment to ensuring payment security for developers and off-takers (distribution companies/bulk consumers) alike, its efficient mechanism has instilled confidence in the RE sector, fostering commercial utilisation and driving down tariffs. The corporation has also ventured into new initiatives such as supply of RE to Commercial and Industrial (C&I) consumers, engaging with relevant stakeholders, expansion of Electric Vehicle (EV) charging infrastructure, and the green hydrogen and ammonia production.

In simpler terms, green hydrogen is a hydrogen gas that's made by splitting water using electricity from renewable sources like wind or solar power. Using power from renewable sources – green hydrogen and green ammonia – is an important part of India's plan to change how we generate and meet energy needs. Sometimes, we have more renewable energy than we need right away; hydrogen lets us save that extra energy for later use. Hydrogen is expected to play a critical role in decarbonisation of industry – fertilisers/refineries in the short run; steel, cement and heavy transport in the long run. SECI is currently looking at about 450000 metric tonnes of green hydrogen production capacity and 1.5 GW of electrolyser manufacturing capacity.These endeavours speak of the organisation's determination to pushing the boundaries of clean energy adoption in India.

SECI'S CONTRIBUTION IN ENERGY TRANSITION

The impact of SECI's initiatives reverberates across the renewable energy ecosystem. With over 60 GW of project capacity awarded through competitive bidding, the establishment has been instrumental in transforming sporadic-level renewable energy projects into commercially viable and mainstream sources of power. More than 20 GW of capacity has already been commissioned, with numerous projects progressing towards implementation.

SECI is a Miniratna (Category-I) Schedule-A Central Public Sector Undertaking (CPSU), and a licensee for trading RE on a pan-India basis. With cumulative PSAs exceeding 51 GW, the CPSU has contributed in shaping and streamlining the RE sector in the country on a substantial scale.

SECI's proactive approach and agility have catalysed a paradigm shift in the energy sector. As RE gains momentum and becomes more competitive in terms of pricing, there's a growing interest in using it as a substitute for traditional energy sources like coal and gas. However, a major challenge with RE is its intermittency - the fact that it's not always available. In the past, it was relatively easy to mix a small amount of renewable energy with a large amount of conventional energy. But as renewable energy becomes a bigger part of our electricity mix (only 12-13% from solar and

